Vaibhav Mahore

AI/ML Engineer | LLM Evaluation • Tool-Using Agents • Efficient Sequence Models

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EDUCATION

Indian Institute of Science (IISc), Bengaluru Aug 2023 - Jul 2027 (expected)
B.Tech in Mathematics and Computing
Coursework: Machine Learning, Deep Learning, NLP, Probability, Linear Algebra, Optimization, DSA
St. Ann's Senior Secondary School (CBSE) Class XII: 2023 | Class X: 2021

EXPERIENCE

AI/ML Research Intern, Ericsson India Bengaluru, May 2026 - Jul 2026

- Benchmarked WiMamba (Mamba wireless model) vs a Transformer on Massive MIMO CSI (AWS A10G): **10.7× lower latency** (30 ms vs 327 ms) and **42.6× less GPU memory** (113 MB vs 4.8 GB) at 4×4 patch size, with linear scaling.
- Evaluated out-of-distribution transfer to an unseen geography (Florida → San Diego) across 5 wireless tasks: **+16.6% composite score** (0.9626) over baseline and **7.4-7.7 dB lower NMSE** on channel estimation and interpolation.
- Parameter-efficient fine-tuning with 75% of parameters frozen retained **98.5%** of full fine-tuning performance.

Machine Learning Engineer, Alignerr and micro1 Remote, May 2025 - Aug 2025

- Alignerr: Wrote reasoning traces and tool-use data for LLM post-training on tool selection and problem solving.
- micro1: Built model evaluation, validation and data-preprocessing workflows in Python (TensorFlow, scikit-learn).

PROJECTS

AIRE: AI Reliability & Evaluation Engine *(Python, RAG, Gemini API, pytest)* Aug - Oct 2026 | GitHub | Report

- Built a trace-based LLM/RAG evaluation framework with a CI regression gate: deterministic metrics (groundedness, citation coverage, Recall@K, MRR), a 6-class failure taxonomy and per-case attribution; 32 tests.
- Compared 3 RAG versions on a 26-case suite; the CI gate blocks a seeded regression where groundedness fell **0.57** → **0.00** even as retrieval recall rose 0.73 → 0.91.
- Evaluated Gemini 2.5 Flash as the generator: abstention **0.0** → **1.0** and groundedness **0.57** → **1.0**, with the same report flagging the cost: 15 s/case latency, 7× tokens and 6/26 rate-limit failures.

Autonomous Repository Engineer *(Python, LLM tool calling, pytest)* Sep - Oct 2026 | GitHub | Report

- Built a coding agent in which the LLM only proposes JSON tool calls: a 9-state machine (plan → act → observe → verify) validates, permission-gates (AUTO/ASK/DENY) and path-jails every call, and finishes only when tests pass.
- Solved 4/4 bug-fix, test-writing and refactoring tasks in scripted mode and 2/2 with a live Gemini 2.5 Flash planner (**0 failed tool calls**); rejected **8/8** path-escape, command-injection and malformed calls; 7 tools, 24 tests.

Efficient Sequence Modeling: S4 to Mamba-3 *(PyTorch, GPU benchmarking)* May - Sep 2026 | GitHub | Report

- Implemented minimal S4D, Mamba, Mamba-2 (chunked SSD) and Mamba-3 in pure PyTorch; 22 tests check that parallel and recurrent forms match (fp32, 1e-3 tolerance), padding invariance and determinism.
- Reproduced the Mamba-3 state-tracking result: **98.3%** parity accuracy with complex transitions vs ≤56.3% (chance) for Mamba-1/2, S4D and a Transformer; selective copying: **67%** (selective SSMs) vs 6% (LTI S4D).
- Benchmarked training from 1K to 32K tokens on a 6 GB RTX 4050: at 16K, S4D's FFT form trained **6.0× faster** per step than a Transformer (0.74 s vs 4.48 s); reproduced the ICML 2024 copying gap (Transformer 100% vs Mamba 0%).

SKILLS

Languages: Python, C++, SQL, Bash
ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face, PEFT, state space models (Mamba, S4D)
LLM Systems: RAG, LLM evaluation and regression testing, tool calling, agent runtimes, Gemini/OpenAI APIs
Tools & Cloud: Git, GitHub Actions (CI), Docker, Linux, pytest, Playwright, AWS (EC2, S3), NumPy, Pandas

ACHIEVEMENTS & CERTIFICATIONS

Amazon ML Challenge: Top 100 national finalist; led Team AutoML. 2026
Oracle Cloud (OCI) Certified Professional: Generative AI (credential) and Data Science (credential) Aug 2025

LEADERSHIP

Sci-Tech Coordinator, Rhapsody Fest, IISc 2023 - 2026

- Ran 3 inter-college gaming tournaments, a coding contest (200+ teams) and a 500+ member Discord community.